

An Introduction to Weka

Named after a flightless New Zealand bird, Weka is a set of machine learning algorithms that can be applied to a data set directly, or called from your own Java code. Weka contains tools for data pre-processing, classification, regression, clustering, association rules, and visualisation.



Machine learning is nothing but a type of artificial intelligence which enables computers to learn the data without help of any explicit programs. Machine learning systems crawl through the data to find the patterns and, when these are found, adjust the program's actions accordingly.

Data mining analyses the data from different perspectives and summarises it into parcels of useful information. The machine learning method is similar to data mining. The difference is that data mining systems extract the data for human comprehension. Data mining uses machine language to find valuable information from large volumes of data.

Weka

Weka is data mining software that uses a collection of machine learning algorithms. These algorithms can be applied directly to the data or called from the Java code.

Weka is a collection of tools for:

- Regression
- Clustering

- Association
- Data pre-processing
- Classification
- Visualisation

The features of Weka are shown in Figure 1.

Installation of Weka

You can download Weka from the official website <http://www.cs.waikato.ac.nz/ml/weka/>.

Execute the following commands at the command prompt to set the Weka environment variable for Java, as follows:

```
setenv WEKAHOME /usr/local/weka/weka-3-8-2
setenv CLASSPATH $WEKAHOME/weka.jar:$CLASSPATH
```

Once the download is completed, run the `exe` file and choose the default set-up.

Weka application interfaces

There are totally five application interfaces available